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Content

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Rating: 5.0

Challenge

[image: The Buggy .Net challenge])

Initial reckoning

The web page shows a POST form that has a single text field called filename:

[image: Main page of Buggy .Net service])

HTTP headers reveal that the server is a Microsoft-IIS/10.0 with [ASP.NET](#) 4.0.30319:

```
HTTP/1.1 200 OK
Cache-Control: private
Content-Type: text/html; charset=utf-8
Server: Microsoft-IIS/10.0
X-AspNet-Version: 4.0.30319
X-Powered-By: [ASP.NET] (https://ASP.NET)
[...]
```

Using "Default.aspx" as the file name in the form retrieves the page source ? ... but wait, the source is available at <http://52.197.162.211/Default.txt> anyways. Also, using "Default.txt" as the file name in the form retrieves that file. And obviously we cannot access the flag at neither "C:\\FLAG.txt" nor "..\\..\\FLAG.txt" ?.

Source code

```
bool isBad = false;
try {
    if ( Request.Form["filename"] != null ) {
        isBad = Request.Form["filename"].Contains("..") == true;
    }
} catch (Exception ex) {

}

try {
    if (!isBad) {
        Response.Write(System.IO.File.ReadAllText(@"C:\inetpub\wwwroot\" +
Request.Form["filename"]));
    }
} catch (Exception ex) {

}
```

Interesting observations

My first thought was something like "wtf? two separate try-catch-blocks?" Can we use that for anything evil? Certainly `isBad` will remain `false` if the first exception handler kicks in.

So how could we raise an exception in the first try-catch-block in order to permit path traversal without raising an exception in the second try-catch-block?

- Can `String.Contains` raise an exception?
- Can `Request.Form[...]` return anything that does not have a `Contains()` member method

but still concatenates as useful string later on?

- Can `Request.Form[...]` raise an exception in some cases (particularly on first invocation) but not in others?

Known bugs in **ASP.NET 4.0** regarding request fields/forms?

So lets ask Google if there are any known bugs in [ASP.NET](#) ...

Interesting, [request validation has changed in \[ASP.NET 4.0\]\(https://docs.microsoft.com/en-us/aspnet/whitepapers/aspnet4/breaking-changes#0.1__Toc256770147\)](https://docs.microsoft.com/en-us/aspnet/whitepapers/aspnet4/breaking-changes#0.1__Toc256770147) (right, that's the version of our attack target!). This seems to have resulted in many complaints!

- [\[stackoverflow.com: A potentially dangerous Request.Form value was detected from the client\]\(https://stackoverflow.com/q/81991/2425802\)](https://stackoverflow.com/q/81991/2425802)
- How to prevent the Exception "A potentially dangerous Request.Form value was detected from the clientside" when Template column is used.)
- [\[ASP.Net Error: A potentially dangerous Request.Form value was detected from the client\]\(https://www.aspsnippets.com/Articles/ASPNet-Error-A-potentially-dangerous-RequestForm-value-was-detected-from-the-client.aspx\)](https://www.aspsnippets.com/Articles/ASPNet-Error-A-potentially-dangerous-RequestForm-value-was-detected-from-the-client.aspx)
- how to try-catch potentially dangerous Request.Form value exception)

That did not help so far ... but then I stumbled upon this: [There's actually someone* telling us that it might not be a good idea to ignore the exception that's thrown on **the first access** to the Request collections.](#)

*) NCC Group is certainly not just someone, but to be honest, I did not really care who's exploit I could recycle back then ?.

Basic idea

The basic idea of that vulnerability is that, for POST requests, request validation prevents "dangerous content" (e.g. HTML tags or similar, such as `

[Original writeup](https://www.sigflag.at/blog/2019/writeup-hitconctf2019-buggy-dot-net/) (<https://www.sigflag.at/blog/2019/writeup-hitconctf2019-buggy-dot-net/>).